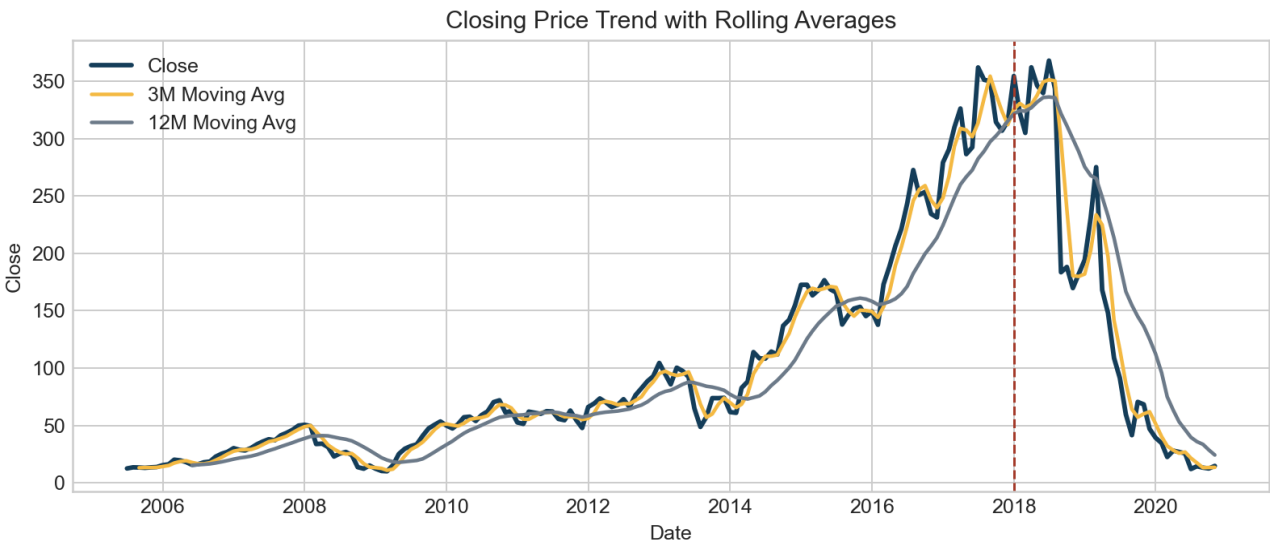


# Yes Bank Stock Closing Price Prediction - Summary

Standalone capstone summary aligned to the reference Yes Bank project, but updated with an original time-aware workflow, expanded EDA, and a Gemini-enabled dashboard experience.

## Project Snapshot

| Metric                          | Value                |
|---------------------------------|----------------------|
| Dataset rows                    | 185                  |
| Date range                      | Jul 2005 to Nov 2020 |
| Peak close                      | 367.90               |
| Best raw correlation with close | Low (0.9954)         |
| Selected deployment model       | Lasso Regression     |
| Final Lasso holdout RMSE        | 16.43                |



## Findings That Stay Close to the Reference Project

- The dataset is clean, with no missing values and no duplicate rows.
- All price variables are positively skewed, and the raw OHLC variables remain highly correlated with closing price.
- The post-2018 period is meaningfully more volatile, so the broader reference story about crisis-era instability still clearly holds.
- Regularized linear models remain strong performers on the same dataset, which keeps the final project direction close to the reference repo.

## Evaluation Summary

| Model                 | MAE   | RMSE  | R2     |
|-----------------------|-------|-------|--------|
| Linear Regression     | 10.08 | 14.48 | 0.9871 |
| Lasso Regression      | 11.05 | 16.43 | 0.9834 |
| Ridge Regression      | 11.48 | 16.83 | 0.9826 |
| ElasticNet Regression | 15.78 | 23.30 | 0.9666 |
| Random Forest         | 28.36 | 36.28 | 0.9189 |
| KNN Regressor         | 58.70 | 74.81 | 0.6554 |

Linear Regression and the regularized linear family dominate the holdout comparison. For deployment, this project chooses Lasso because it stays highly accurate while handling multicollinearity more gracefully than plain linear regression.

### Regime Comparison

| Regime      | Avg_Close | Avg_Range | Mean_Return_Pct | Observations |
|-------------|-----------|-----------|-----------------|--------------|
| 2018 onward | 148.86    | 44.61     | -5.7800         | 35           |
| Before 2018 | 95.02     | 15.68     | 2.6300          | 150          |

The regime table shows why the 2018 break matters: the average monthly range rises sharply and the mean intramonth return turns negative in the later phase.